

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

## ASSESSMENT TOOL 2 – SCENARIO BASED

|                  |                                                                                                                            |               |                                    |
|------------------|----------------------------------------------------------------------------------------------------------------------------|---------------|------------------------------------|
| Course Code:     | DSA03                                                                                                                      | Course Title: | Natural Language Processing        |
| CO Assessed:     | CO2 – Manipulation                                                                                                         | Assessment:   | Assessment Tool 2 – SCENARIO BASED |
| Weightage:       | 30% of CIA                                                                                                                 |               |                                    |
| CO5 Statement:   | Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3) |               |                                    |
| Student Name:    | Shaik Nuha                                                                                                                 | Reg. No.:     | 192472242                          |
| Section / Batch: | B.E-CSE-AI                                                                                                                 | Date:         |                                    |

### Code of Conduct

I \_\_\_\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: \_\_\_\_\_

### Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

| Section / Topic                                                              | Focus                                                                                                                                                                      | Q Nos |
|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| English Word Classes, Tagsets for English, Part of Speech Tagging            | Morphological decomposition of analyzing, analysis, and analytical; identification of roots, affixes, inflectional and derivational forms, and normalization for indexing. | Q1    |
| English Word Classes, Tagsets for English, Stochastic Part of Speech Tagging | Morphological parsing of disagree, agreement, and agreeable; identification of prefixes, suffixes, base forms, and semantic effects of derivation.                         | Q2    |
| Counting Words, English Word Classes, Part of Speech Tagging                 | Morphology-based normalization of govern, government, and governance for topic modeling, clustering, and lexical standardization.                                          | Q3    |

|                                                                                 |                                                                                                                                                   |    |
|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----|
| Rule-Based Part of Speech Tagging, Transformation-Based Tagging                 | Morphological analysis of activate, activation, and reactivation; identification of derivational layers, word-class changes, and semantic shifts. | Q4 |
| Rule-Based Part of Speech Tagging, English Word Classes, Part of Speech Tagging | Inflectional normalization of create, creates, and creating; extraction of root forms and classification of tense/aspect transformations.         | Q5 |

## Annexure A – SCENARIO BASED

1.A large-scale search engine indexing system processes user queries containing word variants such as “analyzing”, “analysis”, and “analytical”. The system must improve retrieval accuracy by grouping semantically related terms under a unified representation. Design a morphological processing approach to decompose each input word into root and affixes, distinguish between inflectional and derivational forms, and normalize them for indexing. Apply the process to the given inputs and determine the root form along with the morphological structure of each word.

2.An AI-based sentiment analysis platform receives product reviews containing words like “disagree”, “agreement”, and “agreeable”. The system must interpret sentiment consistently despite variations in word formation. Develop a morphological parsing strategy to identify prefixes, suffixes, and base forms, while capturing how derivational changes influence meaning. Process the given inputs to extract roots and classify each transformation, ensuring accurate semantic interpretation across all word forms.

3.A text analytics pipeline in a news aggregation system handles documents containing terms such as “govern”, “government”, and “governance”. The system must standardize these variations for topic modeling and clustering. Construct a morphology-based normalization mechanism that decomposes each word into its root and affixes, identifies derivational layers, and maps them to a common base form. Apply the approach to the input words and derive the normalized representation along with their morphological breakdown.

4.A document classification system processes input words such as “activate”, “activation”, and “reactivation” to improve semantic grouping and indexing.

Construct a morphological parsing strategy to identify prefixes, suffixes, and root forms while distinguishing derivational layers.

Analyze how prefix addition and suffix transformation affect meaning and word class.

Apply the process to decompose each word and map them to a unified base representation. Derive the morphological structure and normalized root for each input word.

5. A search optimization module processes input words such as “create”, “creates”, and “creating” to ensure consistent retrieval across grammatical variations. Design a morphology-based normalization approach focusing on inflectional transformations such as tense and aspect.

Identify suffix patterns and map all variations to a common base form without altering meaning.

Apply the process to extract root forms and classify each transformation. Determine the morphological breakdown and normalized output for each word.

| Criteria                     | Weightage (%) | Level 4 (Exemplary)                                                       | Level 3 (Proficient)                               | Level 2 (Developing)                              | Level 1 (Beginning)                               |
|------------------------------|---------------|---------------------------------------------------------------------------|----------------------------------------------------|---------------------------------------------------|---------------------------------------------------|
| Understanding of Scenario    | 20%           | Demonstrates clear and complete understanding of the scenario and context | Shows good understanding with minor gaps           | Partial understanding; misses key aspects         | Misinterprets or fails to understand the scenario |
| Identification of Key Issues | 20%           | Accurately identifies all relevant issues and factors involved            | Identifies most key issues with minor omissions    | Identifies limited issues; some irrelevant points | Unable to identify key issues                     |
| Application of Concepts      | 25%           | Applies appropriate concepts effectively to address the scenario          | Applies concepts with minor errors                 | Limited or partially correct application          | Unable to apply relevant concepts                 |
| Decision Making              | 20%           | Makes well-reasoned, logical decisions with strong rationale              | Makes reasonable decisions with some justification | Makes weak decisions with limited justification   | No clear decisions made or rationale provided     |
| Clarity of Response          | 15%           | Response is clear, structured, and logically presented                    | Generally clear with minor issues                  | Somewhat unclear or poorly structured             | Disorganized and difficult to understand          |

| Q. No. /<br>Criteria<br>Level | Understanding<br>of Scenario | Identification<br>of Key<br>Issues | Application<br>of<br>Concepts | Decision<br>Making | Clarity of<br>Response | Sub.<br>Total | Total<br>% |
|-------------------------------|------------------------------|------------------------------------|-------------------------------|--------------------|------------------------|---------------|------------|
| 1                             |                              |                                    |                               |                    |                        |               |            |
| 2                             |                              |                                    |                               |                    |                        |               |            |
| 3                             |                              |                                    |                               |                    |                        |               |            |
| 4                             |                              |                                    |                               |                    |                        |               |            |
| 5                             |                              |                                    |                               |                    |                        |               |            |

| Faculty In-charge | Course Coordinator | Program Director |
|-------------------|--------------------|------------------|
|                   |                    |                  |
| Signature: _____  | Signature: _____   | Signature: _____ |
| Date: _____       | Date: _____        | Date: _____      |

## 1. Morphological Processing for "analyzing", "analysis", and "analytical"

**Aim:** To design a morphological processing approach that decomposes words into roots and affixes, distinguishes between inflectional and derivational forms, and normalizes them for efficient search engine indexing.

### Step 1: Morphological Decomposition

| Word       | Root    | Affix | Type         |
|------------|---------|-------|--------------|
| analyzing  | analyze | -ing  | Inflectional |
| analysis   | analyze | -sis  | Derivational |
| analytical | analyze | -ical | Derivational |

### Step 2: Morphological Structure

analyzing

analyze + ing

analysis

analyze + sis

analytical

analyze + ical

### Step 3: Classification

| Word       | Morphological Type |
|------------|--------------------|
| analyzing  | Inflectional       |
| analysis   | Derivational       |
| analytical | Derivational       |

#### Step 4: Normalized Form

| Word       | Normalized Root |
|------------|-----------------|
| analyzing  | analyze         |
| analysis   | analyze         |
| analytical | analyze         |

#### Conclusion:

All three words are normalized to the common root analyze, enabling the search engine to retrieve documents regardless of grammatical or derivational variations.

## 2.Morphological Parsing for "disagree", "agreement", and "agreeable"

**Aim:** To develop a morphological parsing strategy that identifies prefixes, suffixes, and base forms while analyzing how derivational changes influence semantic meaning.

#### Step 1: Morphological Decomposition

| Word      | Prefix | Root  | Suffix | Type              |
|-----------|--------|-------|--------|-------------------|
| disagree  | dis-   | agree | —      | Prefix Derivation |
| agreement | —      | agree | -ment  | Derivational      |
| agreeable | —      | agree | -able  | Derivational      |

#### Step 2: Morphological Structure

disagree

dis + agree

agreement

agree + ment

agreeable

agree + able

#### Step 3: Transformation Analysis

| Word      | Meaning Change   |
|-----------|------------------|
| disagree  | Opposite meaning |
| agreement | Noun form        |
| agreeable | Adjective form   |

#### Step 4: Normalized Root

| Word      | Root  |
|-----------|-------|
| disagree  | agree |
| agreement | agree |
| agreeable | agree |

#### Conclusion:

Although prefixes and suffixes modify meaning and grammatical category, all words share the same semantic root agree, improving sentiment analysis consistency.

### 3.Morphology-Based Normalization for "govern", "government", and "governance"

**Aim:** To normalize derivational word forms for topic modeling and document clustering.

#### Step 1: Morphological Decomposition

| Word       | Root   | Affix | Type         |
|------------|--------|-------|--------------|
| govern     | govern | —     | Base Word    |
| government | govern | -ment | Derivational |
| governance | govern | -ance | Derivational |

#### Step 2: Morphological Structure

govern

government

govern + ment

governance

govern + ance

#### Step 3: Derivational Layers

| Word       | Word Class |
|------------|------------|
| govern     | Verb       |
| government | Noun       |
| governance | Noun       |

#### Step 4: Normalized Representation

| Word       | Normalized Root |
|------------|-----------------|
| govern     | govern          |
| government | govern          |
| governance | govern          |

#### Conclusion:

All words are mapped to the root govern, reducing redundancy and improving topic clustering and document indexing.

### 4. Morphological Parsing for "activate", "activation", and "reactivation"

**Aim:** To construct a morphological parsing strategy that identifies prefixes, suffixes, root forms, and derivational layers for semantic grouping.

#### Step 1: Morphological Decomposition

| Word         | Prefix | Root     | Suffix | Type                  |
|--------------|--------|----------|--------|-----------------------|
| activate     | —      | activate | —      | Base Word             |
| activation   | —      | activate | -ion   | Derivational          |
| reactivation | re-    | activate | -ion   | Prefix + Derivational |

#### Step 2: Morphological Structure

activate

activation

activate + ion

reactivation

re + activate + ion

#### Step 3: Effect of Prefix and Suffix

| Word         | Meaning                    |
|--------------|----------------------------|
| activate     | Make active                |
| activation   | Process of becoming active |
| reactivation | Activate again             |

#### Step 4: Normalized Root

| Word         | Root     |
|--------------|----------|
| activate     | activate |
| activation   | activate |
| reactivation | activate |

#### Conclusion:

The prefix re- modifies the meaning by indicating repetition, while the suffix -ion converts the verb into a noun. All forms are normalized to activate.

### 5.Morphology-Based Normalization for "create", "creates", and "creating"

**Aim:** To normalize inflectional variations of words for consistent search optimization and indexing.

#### Step 1: Morphological Decomposition

| Word     | Root   | Suffix | Type         |
|----------|--------|--------|--------------|
| create   | create | —      | Base Word    |
| creates  | create | -s     | Inflectional |
| creating | create | -ing   | Inflectional |

#### Step 2: Morphological Structure

create

creates

create + s

creating

create + ing

#### Step 3: Inflectional Classification

| Word     | Transformation        |
|----------|-----------------------|
| create   | Base Form             |
| creates  | Third Person Singular |
| creating | Present Participle    |



#### Step 4: Normalized Output

| Word     | Root   |
|----------|--------|
| create   | create |
| creates  | create |
| creating | create |

#### Conclusion:

The suffixes -s and -ing represent grammatical changes without altering the meaning. All forms are normalized to the common root create, improving search optimization and information retrieval.